

NOVOKUZNETSK, Russian Federation

WMO: 298460

Lat: 53.817N

Lon: 86.883E

Elev: 308

StdP: 97.68

Time Zone: 7.00 (E07)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.6	-36.0	0.1	-33.2	-33.1	0.2	-30.4	14.2	-6.3	12.3	-6.2	1.2	340	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.5	19.4	27.7	18.6	26.0	18.0	20.9	26.9	19.8	25.5	18.9	24.3	3.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.3	23.5	17.8	13.3	22.5	16.9	12.5	21.7	62.1	26.8	58.4	25.5	55.2	24.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.5	(b) 9.5	(c) 8.2	(e) -35.0	(f) 33.0	(g) 4.7	(h) 1.8	(i) -38.4	(j) 34.2	(k) -41.2	(l) 35.2	(m) -43.8	(n) 36.2	(o) -47.3	(p) 37.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-16.5	-12.7	-4.9	3.9	11.4	17.0	19.0	16.6	10.3	3.4	-5.8	-13.1
	DBStd	13.49	8.12	7.11	6.14	5.56	5.15	4.00	3.14	3.63	4.26	5.26	6.90	7.72
	HDD10.0	3612	821	636	463	193	47	3	0	1	48	211	474	715
	HDD18.3	5902	1079	869	721	433	220	69	31	77	244	462	724	973
	CDD10.0	861	0	0	0	11	91	213	279	206	56	7	0	0
	CDD18.3	109	0	0	0	0	5	29	51	22	2	0	0	0
	CDH23.3	1109	0	0	0	2	106	306	436	225	35	0	0	0
	CDH26.7	266	0	0	0	0	23	69	108	59	7	0	0	0
Wind	WSAvg	3.2	2.9	3.2	3.6	3.8	3.7	2.8	2.4	2.5	2.8	3.5	3.7	3.5
Precipitation	PrecAvg	714	43	32	42	46	59	75	84	73	58	58	79	65
	PrecMax	894	73	65	114	104	131	140	128	108	133	130	134	124
	PrecMin	569	8	5	9	13	29	41	39	33	21	30	32	23
	PrecStd	79	18	14	24	22	25	23	24	23	22	22	24	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	2.7	10.9	22.2	29.6	31.0	31.5	31.9	27.2	20.1	10.0	2.6
		MCWB	0.8	0.2	5.5	11.4	17.7	18.7	20.2	21.0	16.5	11.4	6.2	0.7
	2%	DB	-0.7	0.4	7.2	19.1	26.2	28.6	29.3	28.1	23.2	16.2	7.3	0.2
		MCWB	-1.4	-1.2	3.1	10.5	15.5	18.3	20.0	19.1	14.9	9.8	4.8	-0.5
	5%	DB	-3.1	-1.1	5.0	16.1	23.9	26.8	27.7	25.9	20.2	13.8	4.2	-1.7
		MCWB	-4.0	-2.5	1.9	9.0	14.3	17.9	19.4	18.3	13.8	8.8	2.5	-2.7
	10%	DB	-5.4	-3.0	3.0	13.0	21.1	24.9	25.9	23.8	17.8	11.0	2.2	-3.1
		MCWB	-6.2	-4.1	0.7	7.4	12.9	17.1	18.9	17.2	12.3	7.1	0.9	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.9	5.8	12.7	18.8	20.9	22.8	22.1	18.2	12.2	7.1	0.9
		MCDB	1.4	1.9	10.2	19.9	27.3	26.6	28.4	29.0	24.6	18.0	9.4	1.8
	2%	WB	-1.5	-1.0	3.5	11.1	16.7	19.7	21.3	20.2	15.7	10.3	4.7	-0.4
		MCDB	-0.8	0.1	6.4	17.0	24.1	26.2	26.9	26.6	21.8	15.6	7.3	0.2
	5%	WB	-4.1	-2.6	2.2	9.6	15.0	18.7	20.3	19.0	14.2	9.0	2.7	-2.4
		MCDB	-3.2	-1.3	4.6	15.4	21.8	25.0	25.6	24.5	19.2	13.2	4.3	-1.8
	10%	WB	-6.3	-4.2	0.8	7.9	13.6	17.7	19.4	17.8	12.9	7.5	1.0	-4.0
		MCDB	-5.5	-3.1	2.7	12.5	19.7	23.4	24.3	22.9	16.8	10.6	2.0	-3.1
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.9	9.3	10.4	11.9	11.1	10.5	10.7	10.6	8.6	6.6	7.2
		MCDBR	7.9	8.6	10.6	15.6	16.4	14.5	13.5	14.5	15.8	13.6	7.7	8.1
		MCWBR	7.6	7.5	7.5	8.7	8.4	6.7	6.0	6.7	8.0	7.8	5.9	7.5
	5% WB	MCDBR	7.8	8.7	9.7	14.3	14.4	12.5	11.8	12.9	14.0	12.8	7.5	8.0
		MCWBR	7.6	7.8	7.0	8.7	8.1	6.6	6.2	6.5	7.5	7.8	5.9	7.7
Clear-Sky Solar Irradiance	taub	0.297	0.336	0.386	0.420	0.417	0.413	0.430	0.413	0.376	0.365	0.344	0.309	
	taud	1.998	1.937	1.873	2.023	2.176	2.272	2.237	2.263	2.308	2.201	1.997	1.936	
	Ebn at Noon	612	712	760	799	829	837	813	804	787	688	537	492	
	Edh at Noon	77	115	153	151	137	126	128	118	99	88	77	68	
All-Sky Solar Radiation	RadAvg	0.99	2.02	3.34	4.33	5.18	5.97	5.49	4.57	3.14	1.62	0.80	0.62	
	RadStd	0.11	0.17	0.36	0.31	0.64	0.47	0.36	0.35	0.36	0.19	0.14	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-3.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page